Quantitative assessment of T-cell repertoire recovery after hematopoietic stem cell transplantation

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Evidence abstract

Delayed T cell recovery and restricted T cell receptor (TCR) diversity after allogeneic hematopoietic stem cell transplantation (allo-HSCT) are associated with increased risks of infection and cancer relapse.

Technical challenges have limited faithful measurement of TCR diversity after allo-HSCT.

Here we combined 5' rapid amplification of complementary DNA ends PCR with deep sequencing to quantify TCR diversity in 28 recipients of allo-HSCT using a single oligonucleotide pair.

Analysis of duplicate blood samples confirmed that we accurately determined the frequency of individual TCRs.

After 6 months, cord blood-graft recipients approximated the TCR diversity of healthy individuals, whereas recipients of T cell-depleted peripheral-blood stem cell grafts had 28-fold and 14-fold lower CD4(+) and CD8(+) T cell diversities, respectively.

After 12 months, these deficiencies had improved for the CD4(+) but not the CD8(+) T cell compartment.

Overall, this method provides unprecedented views of T cell repertoire recovery after allo-HSCT and may identify patients at high risk of infection or relapse.

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